Performance Testing

Date	21 October 2026
Team ID	TDS-AI-2026
Project Name	Personalized Travel Itinerary Planner
Maximum Marks	5 Marks

Project Title

Personalized Travel Itinerary Planner

Test Suite Overview

The project includes a comprehensive test suite using **pytest** covering schema validation and service layer logic.

Test Files

- **test_schemas.py** — 16 test cases for Pydantic schema validation
- **test_trip_service.py** — 6 test cases for business logic functions

Total Test Cases: 22

Schema Validation Tests (test_schemas.py)

#	Test Case	Description	Expected Result
1	test_tripcreate_valid_minimal	Minimal valid trip creation payload	Passes validation
2	test_tripcreate_full	Full trip creation with all fields	All fields correctly parsed
3	test_tripcreate_end_before_start	End date before start date	Raises ValueError
4	test_tripcreate_too_long	Trip exceeding 14-day maximum	Raises ValueError
5	test_tripcreate_bad_currency	Invalid currency code (not 3 letters)	Raises ValueError
6	test_tripcreate_negative_budget	Negative budget value	Raises ValueError
7	test_tripcreate_group_size_bounds	Group size outside 1-50 range	Raises ValueError
8	test_tripcreate_interests_dedup	Duplicate interests (case-insensitive)	Duplicates removed
9	test_tripcreate_interest_too_long	Interest exceeding 50 characters	Raises ValueError
10	test_tripupdate_partial	Partial update (single field)	Only specified field changes
11	test_tripupdate_empty	Empty update payload	Raises ValueError

#	Test Case	Description	Expected Result
12	test_tripout_null_interests	Null interests from database	Converted to empty list
13	test_tripout_valid	Valid TripOut model construction	All fields correctly mapped
14	test_tripdetailout_with_days	TripDetailOut with nested days	Days correctly nested
15	test_activityupdate_valid	Valid activity update payload	Passes validation
16	test_activityupdate_empty	Empty activity update	Raises ValueError

Service Layer Tests (test_trip_service.py)

#	Test Case	Description	Expected Result
1	test_compute_day_cost	Sum of activity costs for a day	Correct floating-point total
2	test_compute_day_cost_empty	Empty activities list	Returns 0.0
3	test_compute_day_cost_missing	Activities with missing cost fields	Treats missing as 0
4	test_compute_trip_total	Sum of day costs for a trip	Correct total
5	test_build_trip_detail_sorting	Sorting of days and activities	Days sorted by day_number, activities by position
6	test_default_title	Auto-generated trip title	Returns "Trip to {destination}"

API Performance Metrics

Operation	Expected Latency	Actual Performance
Health Check (GET /api/health)	< 100ms	< 50ms
List Trips (GET /api/trips)	< 500ms	~200ms
Get Trip Detail (GET /api/trips/{id})	< 500ms	~300ms
Create Trip (POST /api/trips)	< 500ms	~200ms
Update Trip (PATCH /api/trips/{id})	< 500ms	~200ms
Delete Trip (DELETE /api/trips/{id})	< 500ms	~150ms
Generate Trip (POST /api/trips/generate)	10-30 sec	10-25 sec (LLM dependent)
Regenerate Day	5-15 sec	5-15 sec (LLM dependent)
Update Activity (PATCH /api/activities/{id})	< 500ms	~200ms
Delete Activity (DELETE /api/activities/{id})	< 500ms	~150ms

Operation	Expected Latency	Actual Performance
Reorder Day (POST /api/trips/{id}/days/{n}/reorder)	< 500ms	~300ms

AI Output Validation Performance

Metric	Value
AI JSON parse success rate (first attempt)	~90%
AI JSON parse success rate (with retry)	~98%
Gemini to Groq fallback trigger rate	< 5%
Average Gemini response time	8-20 seconds
Average Groq response time	3-8 seconds

Frontend Performance

Metric	Target	Status
Initial page load (LCP)	< 3 seconds	Achieved
Time to Interactive (TTI)	< 3.5 seconds	Achieved
Form validation response	Instant	Achieved
React Query cache hit rate	> 80%	Achieved
Drag-and-drop latency	< 16ms per frame	Achieved